

Acquiring the History of the City with Collaborative Game Based Learning

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Abstract. The paper presents the 3D maze collaborative game for learning the history of the city. The idea of the game was described as also the main functionalities. Collective gameplay consists in helping the lost participant to get between two points of the 3D maze. The research on the influence of joint gameplay on historical facts acquiring was made. The results of interview and survey were described. The study focused on game design and gamebased learning results.

Keywords: Game based learning · Collaborative gaming · History

1 Introduction

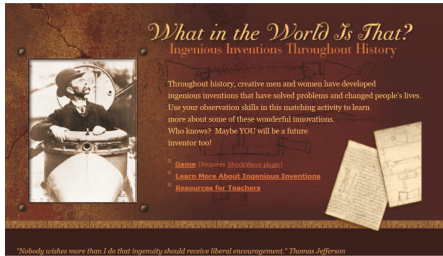
The most effective learning, deepened and permanent, takes place when is initiated and continued by the learner [1]. To make this possible, we need to create a learning environment, an indispensable space of freedom and an autonomous, creative action where digital natives have a chance to show their strengths and develop their potential. Teaching history, especially the home town history is a difficult task [2].

On the Internet, one can find a lot of games that are created solely for entertaining purposes, taking valuable time to the players who could devote themselves to developing their knowledge or skills [3]. One of the best ways to learn is learning through fun, games are motivating, accelerate user response time and simulate real situations. In decision-making games, the primary task is not simply to provide entertainment, but to solve a problem or goal [4].

There are many historical games where the players can acquire the knowledge about the history. For games of this type can be included among others: What in the World is That?, Gladiator, The Battle of Waterloo, Iron Age Life, Vikings, Mummy Maker, Battlefield Academy. What is unique about these games is that the gameplay is one-player and the knowledge is acquired by solving the given tasks (Fig. 1a).

Cooperative games focus on promoting collaboration, learning work in a team and making common decisions [5]. These are autonomous games in which entertainment is only a subordinate goal. They offer an effective approach to learning

and developing skills [6]. These games combine elements of teaching, training, communication and information with entertainment elements [7]. Cooperative games are designed to achieve tangible, lasting changes in the behaviour of their users [8]. The sample of such game is “Keep talking and nobody explodes”, where one player is trapped in a virtual room with the bomb. The other players, which cannot see the bomb, have to help him to defuse the bomb using defusal manual and information given by trapped user (Fig. 1b).



(a)



(b)

Fig. 1. Sample games: (a) historical “What in the World is That?” [9], (b) cooperative “Keep talking and nobody explode” [10]

There are many games in the network that use similar mechanics and the way of players connection, to that presented in the article. However, one can not find an identical solution that uses the world as a 3D maze game with quests and enables players to cooperate. This makes the described in article solution innovative.

In this paper the authors studied the history teaching path with the use of collaborative 3D maze game. The results show the influence on gathering the historical facts about the home town with the use of designed serious game. The remainder of the paper is organised as follows: Sect. 2 describes the game idea and the way of building. Section 3 describes the research way in the form of interviews and survey. Results were also presented in this chapter. Conclusions are presented in Sect. 4.

2 The Game

When designing the games for history it is important to consider the pedagogical goals [11]. The teacher can focus more on historical facts and data or (and) resource management and decision making [12]. The balance between the retaining historical accuracy and gameplay has to be considered. Well designed team games allow students to effectively assimilate historical facts and understand the atmosphere of history [13].

According to the definition the labyrinth is a complicated route with many passages, which is virtually impossible to navigate without a guide [14].

In ancient times, the maze served as protection against unauthorized persons entering the tomb or vault. Today is used in pop culture or as a tourist attraction. The form of labyrinth (3D maze) that has been used in the game draws direct inspiration from the book “Harry Potter and the Goblet of Fire” by Rowling [15]. In this book, the exit from the labyrinth does not directly relate to finding a path that leads to the way out, but to the particular point where the artefact is located. This artefact is symbolizing the successful completion of the quest.

Learning situation, in which the described game can be used is that, the teacher wants to learn young people some historical facts about home city or like in presented paper to gather more historical knowledge because of Lublin citys 700th anniversary. The anniversary is an good opportunity for city residents to play cooperative web game to learn more facts from the history of the city. The content that is useful for learners are unknown home city historical facts gathered through quests and riddles which are randomly picked up from database. In described case the teaching equipment and learning environment are PC with web browser and 3D maze collaborative game.

The game is a turn-based, collaborative team game between its participants. Players, in order to achieve the goal and win, should help each other. The obvious goal of the game is to get through the maze as quickly as possible from the starting point to the destination and to find a trophy cup. The hidden objective of the game is to learn the history of the city of Lublin. For this purpose, in the maze there are riddles/quests that the player must solve. Riddles are related to the history of the city. The player should correctly solve all the quests - only then the trophy cup will appear somewhere in the maze and you will be able to finish the game successfully. The game will end itself (without the trophy-cup, that is with the loss) after a certain time. Such limitation makes the game more dynamic.

The cooperation of many (at least two) players was organized as follows. Exactly one player (called Lost) is supposed to move around the maze and solve riddles/quests. This player is randomly selected from the group of players who have signed up for the game. The other players (called Helpmates) are supposed to suggest the Lost player the solution. They can communicate with each other, as well as receive a quest from Lost player (in the form of scene picture which Lost see), and send suggestions to him (or consult each other). Communication between the Lost and Helpmates players is limited by the time (type: not more than) and by the number (max. 50 during the game). The victory and time of its achievement in the game is assigned to all its participants and is the basis for the statistics of players.

An essential element of the game, apart from creating a situation involving cooperation between players, is its educational element. It was made using the riddles/quests depicted on the walls of the maze. These quests concern the history of the city of Lublin. They are randomly drawn from a large quests database. Four types of riddles/quests and their graphical interfaces are used in the game:

- Indicate appropriate colors (e.g. city coat colors).

- Enter the date (e.g. the date of signature of the Polish-Lithuanian union in Lublin).
- Indication of the correct answer (e.g. choosing one of the 3 answers to the question).
- Pressing the buttons in the specified order (this type is used for scheduling, e.g. serialized during an event).

After giving the correct answer, Lost is rewarded, and in the opposite case, riddle/quest is reappearing. Image of the quest interface the Lost player can send to the Helpmates for consultation.

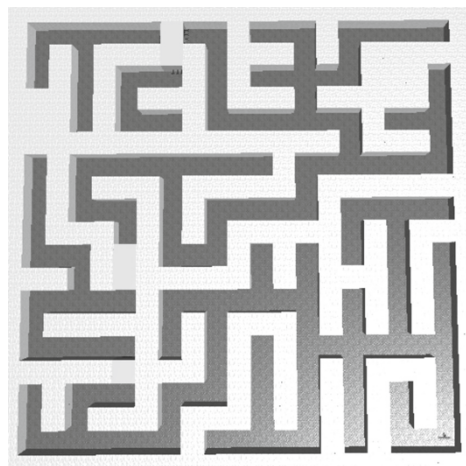


Fig. 2. Sample look of generated 3D maze



Fig. 3. Sample maze scenes with quests and message

The 3D maze is generated randomly in the created game room. At the stage of game development that was possible to look at the maze from above. The sample top look of 3D maze is shown in Fig. 2. During the game the communication between Lost player and helping team is carried out with the use of short text messages appearing on the screen (Fig. 3).

2.1 Functional and Non-functional Requirements of the Game

The main functional requirements for the web technology based 3D maze game are:

- Anyone can sign up for the service.
- The logged in users can edit their account, create or join existing game rooms.
- When creating a new game room, user gives the room name, the number of players, the game time and difficulty, and the optional password. Once created, the creator is the administrator of the room.
- An administrator of the room can discard or block a player, delegate administration to another player, and edit room options.
- Roles of players are awarded randomly.
- Players are communicating using text messages.

Non-functional requirements of the game can be described as follows:

- You can take part in one game at a time and join one room at the same time.
- The user can not block other users' accounts and suspend the game.
- An unregistered person can not participate in the game.
- Play time is limited to the following three options: 5, 10 and 15 min.
- There must be a minimum of two players per room.
- The person using the site must have a browser supporting JavaScript and minimal requirements are: Chrome version 30, Firefox version 30, Internet Explorer 11, Opera version 12.10 or later,
- The server on which the site will be hosted should have a PHP 5.5 or later interpreter and a PostgreSQL 9.3 or later server.

2.2 Data Model and Game Logic Design

The game consist of several quests. First of all the lost player needs to find the all places in the labyrinth to solve the riddles. Whole data structure of the game is presented in the Fig. 4.

It can be noticed that all riddles/quests are stored in the tables: *puzzle*, *puzzle_group* and *puzzle_element*. The riddles (quests) are divided into the four main groups and several subgroups presented in Table 1. Each question in the database has assigned difficulty level which is used during room creation.

The main idea of the functionality of the game application is presented in Fig. 5 in the form of a use case diagram. This diagram shows the roles of the application users and the basic use cases. The application user roles (any person becomes user as a result of the application registration process) were defined as follows (actors shown in Fig. 5):

- Player - any person who registered in the web application.
- Game Administrator - a player who initiated another (named) turn of the game, resulting in a creating maze, starting and ending points, a collection of riddles-quests, difficulty of the game and other gameplay parameters.

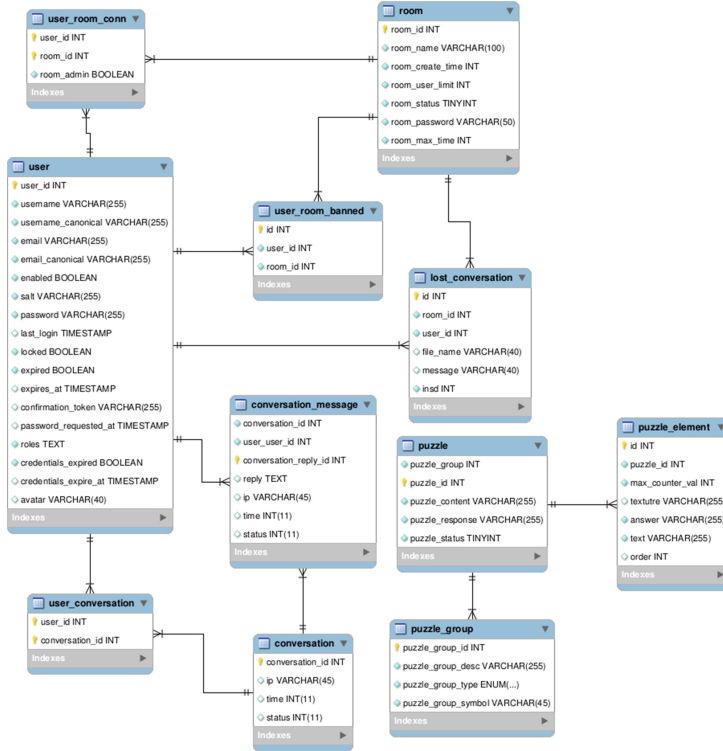


Fig. 4. Game data structure in the form of ERD diagram

- Lost - a player who moves through a virtual maze in a given game.
- Helpmate - a player or their group who works with Lost and prompts him to solve riddles/quests.
- Time - an actor who maps the passage of time (e.g., to end the game after the elapsed time specified by the Game Administrator).

Figure 5 (use cases) describe the main functionalities of the application. These are:

- Edit account data - manage your player profile data.
- Create game room - allows the Player to create a new game, in which the Player is the Game Administrator. The application generates a new maze, sets the start and end points, and generates (gets from the Database) and arranges riddles-quests in the maze.
- Join the game - gives you the ability to join a specific (created and named) game as a participant. The application randomly assigns the role of participant (Lost or Helpmate).
- Manage the game - allows the Game Administrator to set its parameters (e.g. duration), manage players (e.g., remove a player from the game) and start the game.

Table 1. Types of quests

Riddle/quest type	Subtype	Number of quests
Set order	Universities	3
	Districts	4
	Historical facts	3
	Annexation	2
Set date	City	2
	World War II	2
Set colours	Lublin City	2
	Voivodeship	2
Answer select	Monuments	2
	Local authorities	2

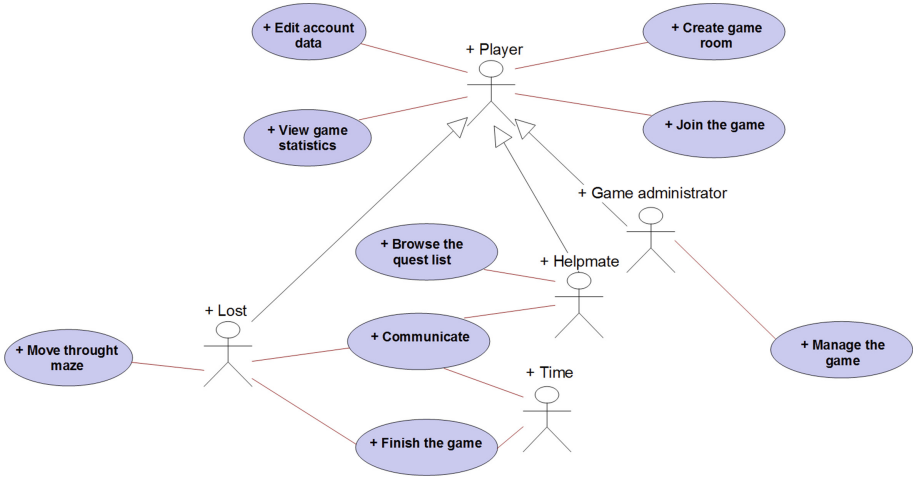


Fig. 5. Use case diagram of the application

- Move through maze - change the position in the maze by Lost and answer the riddles-quest. While moving, Lost has the ability to communicate with Helpmates.
- Communicate - allows communication over the network (typical chat) between Lost and Helpmates and between Helpmates. Communication is limited by the time (Time actor) and quantitative.
- Browse the quest list - browse through the list of riddles/quests to identify an extended description of the riddle/quest. Helpmate gets from Lost the look of the riddle/quest and needs to refine it based on a list of riddles-quests, to work out in group answers sent to Lost player.

- Finish the game - finish the game and memorize its outcome. Completion of the game can be successful (Lost finds the cup) or without (passage of time).
- View game statistics - view statistics of past games, including the top players list.

3 Research Results

In this study a series of interviews with current students were performed. They were students of computer science who had completed the education of history at the secondary school level. During the interviews the students were examined with their reference to games, their motivations for playing games, their attitude to game based learning and knowledge about historical facts of home town. After playing the cooperative game the students filled out a survey questionnaire. The 30 postgraduate students were examined, with the intention of how they acquired the new historical facts about the home city, which they did not know before, with the use of 3D maze collaborative game.

The series interviews were conducted with students, who were known to the researchers. There were 21 male and 9 female participants, with the ages of 21–23 years. Seventeen persons were people who considered themselves as a playing games in their leisure time and the rest of them said, that they prefer the other type of amusement.

A short survey of ten questions was designed to examine the gaming preferences of the students and their attitudes in the collaborative gaming. Interviewees were asked whether they played computer collaborative games often, regularly or never. The next question was about the experience of playing collaborative game for learning and feeling positively motivated, not motivated or negatively motivated. At this stage of the research this three-point scale was sufficient to differentiate between positive and negative perceptions.

In the survey there was also a question how the students prefer to learn and acquire new knowledge, independently, in 2–3 persons group or group over 3 persons. The rest of the questions in the survey concerned the knowledge gained during the game.

The interviews showed that the participants who considered themselves to be game players had different motivations. Among these students the primary motivations for playing games were mental challenge and social experience. In the case of people who did not consider themselves gamers they could play games in the case of boredom or social events. However most students (23) said that acquiring historical knowledge with the use of 3D maze game could be interesting.

The results of the survey showed, that the 40% of the students played the collaborative games often, over 53% who said they were playing regularly and almost 7% who said that never played. It can be seen that in this group over 93% of interviewees played in the collaborative games. Although only 34% of the group have ever come across earlier with game based learning. The results of being motivated were as follows: over 73% were positively motivated, almost 17% were not motivated and 10% was negatively motivated.

The question about the acquiring new knowledge in group gave an interesting results, 54% of students prefer to learn independently, 33% want to learn in 2–3 persons groups and 13% like to learn in larger group. This shown an interesting trend that students are more likely to cooperate in the group and acquire new knowledge when playing the game.

An interesting results were obtained with the questions about historical facts absorbed during the 3D maze game. In the case of “Set colours” quest type 70% of students memorised the proper order, however the “Set date” quests were memorized only by almost 27% of students. In the “Set order” riddle type most problems raised questions about historical facts. Only almost 17% of students remembered the proper order of historical facts after playing the game. However in the case of other subtypes of “Set order” type the results were much better. Average percent in these subtypes was over 51%. “Answer select” riddle type gave an 40% result in acquiring the new knowledge.

4 Conclusions

Historical space since the birth of the games has been the background for their scenarios, and the presence of characters, events or historical symbols has become inherent attribute of the games. Historical games are characterized by, on the one hand, attempts at a realistic, mimetic representation of the past, and on the other, allow for a fictional story. Realism involves the need for consistency with the knowledge of the structure of the past presented and the experience of the players.

It should be noted that in historical games one can talk about the disappearance of the distinction between fiction and reality, the real world and the virtual world. The image of the past created in the game is not an imitation, representation or representation of past reality - it is a cultural construct, attractive primarily because its creator is a player.

At the same time it is important to emphasize that historical games are not only creative tools, but they also require knowledge or familiarity of historical realities. Some of them try to remain faithful to factographies and are educational or educational aided games - or, alternatively, to counter the current knowledge of historical decision, remain faithful to the historical image of the epoch scenery (e.g. Europa Universalis) [16].

The results of the research clearly showed that the students can acquire the home town historical facts much faster when they are playing collaborative game. However there were persons which were not motivated by the game based learning. An immersive game environment in the form of 3D maze connected with the collaborative game enabled players to learn some knowledge during gameplay.

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